Detection of Economic Denial of Sustainability using Time Spent on a Web Page in Cloud

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***Abstract*—Cloud computing is a form of utility computing where the consumers leverage computing resources. Billing is done on an hourly basis based on the amount of resources consumed. In the case of web servers the cost is based on the amount of data that leaves the web server and the number of HTTP requests received.The cloud service provider automatically scales the resources when there is a hike in consumers traffic and the cost on the consumer increases accordingly. This model can put cloud at risk by a tactful attacker who intends to lessen the profit and increase the cost for a consumer. The attacker synthetically generates HTTP requests to the web application so that they are accounted for billing but are not profitable to application provider. Thus, causing a sustainable decline in the economy of the consumer, technically named as Economic Denial of Sustainability(EDoS). The http traffic generated by the attacker mimics the genuine traffic in order to go undetected. This paper proposes a detection mechanism based on the Time Spent on a Web Page(TSP) defined as time spent on viewing a web page. The TSP of the attack traffic differs from the mean TSP of a web page. This deviation of TSP from the mean is calculated taking the exponential distribution of the TSPs and the calculated value is used to detect the surreptitious behavior.**

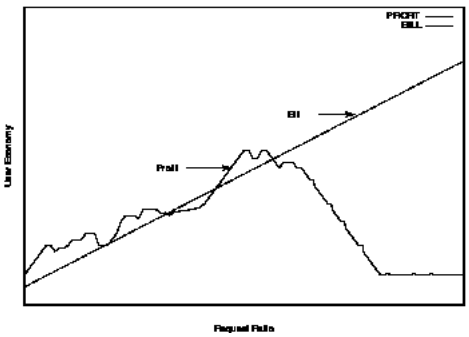


Fig. 1. Variation in Cost and Profit with Request Ratio

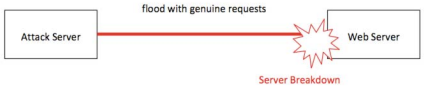


Fig. 2. EDoS attack on non-cloud platform

I. Introduction

Cloud Service Provider(CSP) gives the computing resources on lease to its consumers based on an agreed pricing model.The Amazon EC2 offers, web servers to host their web applications and billing is done based on an hourly basis. The amount of data that leaves the web server along with the number of requests received are considered for billing. The cloud consumer earns profit from a successful transaction on a web application such as a purchase on an ecommerce application [1]. If the user requests are made only for browsing the site and the purchases are never made, the consumer earns no profit but in turn pays to the CSP. Therefore the requests can be classified as profitable and unprofitable to the consumer.

If an attacker intentionally tries to increase the unprofitable to profitable ratio(Request Ratio) the consumer ends up in earning less and paying more. The decline in the profit curve and the stability in the cost curve with increase in the Request Ratio is shown in Figure 1. Initially the profit has minor fluctuations due to the presence of unprofitable requests from genuine users, later the profit decreases due to increase in the attacker's unprofitable requests. A consumer expects a sustainable profit after hosting his application on cloud, EDoS attack denies the customer of this sustainability by increasing the cost and decreasing the profit. The inverse relationship followed by cost and profit during the attack is termed as Economic Denial of Sustainability(EDoS).

Auto scalability feature of cloud makes EDoS possible on a cloud platform. In case of a non-cloud platform, if an attacker tries to send considerable amount of fake traffic, there is a possibility that the target web server fails in serving the requests and becomes unavailable whereas, in cloud this may be misinterpreted as high genuine traffic leading to automatic scaling of resources(virtual instances). The scenario discussed above is shown in Figure 2 and Figure 3. This puts forward the need to identify the attack traffic and bill only for genuine traffic of a web application.

The attacker generates the requests by forming a distributed botnet on internet. The bots can be programmed to generate requests at a very high rates as in the case of a DDoS attack or can generate requests at moderate rate similar to a flash crowd attack [2]. The consumer experiences high costs in less period of time for a DDoS attack whereas, the cost gradually increases for a flash crowd attack.

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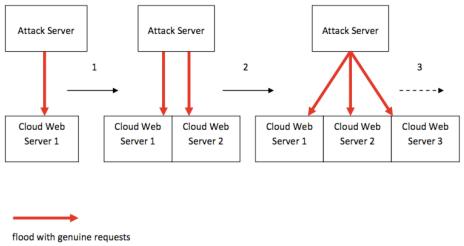


Fig. 3. EDoS attack on cloud platform

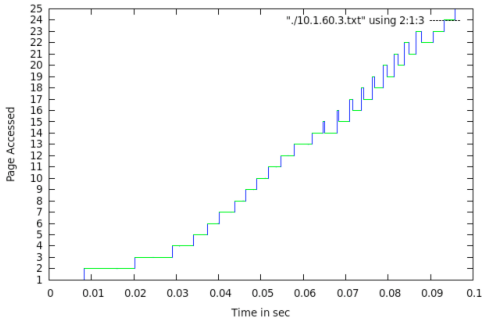


Fig. 4. Distribution graph for http requests

In both cases the requests mimic a genuine request making it difficult to detect but request dynamics and request semantics can be used to differentiate the human generated and bot generated requests [3].This paper considers the Time Spent on a Web Page(TSP) as a request dynamic to detect the attack traffic. If the traffic is illegal the TSP's of the web pages requested deviate from the mean TSP of the web page. The deviation of the TSP from the mean value is used to differentiate the traffic.

II. Background

A. Actors in cloud

The following actors as described in [4] attribute to a successful EDoS attack.

- Cloud Service Provider(CSP) who rents the computing resources to the public and has control over the billing activity.
- Cloud Consumer who purchases the resources offered by the CSP. In this papers context resources refer to web servers required to host the web applications of the cloud consumer.
- Client who accesses the services of the web applications in a legitimate manner.
- Attacker who intentionally generates fraudulent traffic to hit the economy of cloud consumer.

B. Time spent on a page(TSP)

TSP is one among the major factors that determines the quality of a web page. High TSP value may indicate the complex nature and less time may indicate the presence of poor and uninteresting content of a web page. Apart from the quality of web pages TSP can also be used to detect the bot behavior from human behavior. In practice, a huge volume of very small TSP values indicates robot visits [5]. Each web request of each user is stored in the form of user logs. The logs are preprocessed to calculate the TSP for each page.

C. Foot step graphs

In order to analyse the user behavior on a web site the click stream data is collected and represented in the form of a footstep map. If the browsing activity of the user becomes too complex it becomes difficult to analyse the footstep map. [6]provides an easier way to represent the browsing activity, which is called as the footstep graph. These graphs also represent the TSP value for each request of the web page and hence can be used to view the TSP variation visually.

III. Related Work

Christofer Hoff in 2008 coined the term EDoS and describes the fact that the utility computing model and the scalability concepts of cloud computing can be put into risk [1]. The attack is defended using a reverse-turing test called CAPTCHA and a blacklist policy in [6]. When a client requests for a web page it is challenged back with a client puzzle in the form of CAPTCHA and if the client is able to solve,he is allowed access to the site. This mechanism detects the bots as they cannot solve CAPTCHA puzzles like humans. In reality, reverse turing tests are not encouraged as they annoy the clients accessing web application.

A similar type of attack called Fraudulent Resource Consumption(FRC) attack is discussed in [4] and is detected by applying Zipfs law and entropy detection strategies. If the data set under concern violates the laws then an anomaly is detected. [3] puts forward techniques like request dynamics, request semantics and deception to detect flash crowds which can also be adopted to deal with EDoS attack.

IV. Attack Analysis

In order to create EDoS attack the obvious choice for an attacker is to form a botnet and target the victim.The attacker has to program the bot for generating http requests for the web pages. If the possibilities of programming the bot to automate the generation of requests are enumerated, the following possibilities occur.

- If the attackers intention is to generate requests at very high rates and in short period as in the case of a DDoS bot it can be observed that the TSP on the requested page is nearer to zero.
- If the intention is to generate at a moderate rate so as to blend the traffic into the genuine traffic such as flash crowds, the TSP can vary either periodically with equal TSP values or randomly with a range of TSP values.

A sample attack traffic represented in the form of a footstep graph is as shown in Figure 4. The horizontal lines indicate the TSP of a page (the Transition Time from one page to another) and the vertical line represents a transition to the next page.The probability distribution of page requests follows poisson distribution whereas the TSPs represent an exponential distribution [7].

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V. Proposed System

The TSPs that are generated by the bot deviates from the average TSPs that are observed for genuine traffic even though the bot is well programmed. It is not feasible for the attacker to predict the average TSP of a page on the target web site and program accordingly.

The TSP of the attacker's requests deviates largely from the mea n of genuine requests. Therefore the Mean Absolute Deviation(MAD) of TSP becomes a considerable factor to differentiate the illegal traffic.

A. Calculation of TSP

The calculation of TSP is done as mentioned in [5]. Let S_i represent the ordered sequence of the form $< P_{ij}, T_{ij} >$ where P_{ij} represents the page accessed and T_{ij} represents the time stamp of the request for j^{th} page of ith user.

If T_{ij} and T_{ij+1} for two consecutive requests in the set S_i then

$$TSP(P_{ij}) = T_{ij+1} - T_{ij} \quad (1)$$

B. Calculation of MAD

Let N be the number of pages in the web site. Let X_i denote the data set representing the page requests for a page $i \in N$ and $m(X_i)$ denotes the average of TSPs for all requests in the set X_i.

$$m(X_i) = 1/n \sum_{j=0}^{n} TSP(X_{ij}) \text{ where n = cardiinality of } X_i \quad (2)$$

The requests generated by each cloud user is collected in form of logs at cloud controller. Each user has a log represented as L is then analyzed to calculate the MAD exhibited by the requests generated.

$$\forall i \in \text{ N }, L_i \text{ represents the log requests for the page i} \quad (3)$$

$$\forall j \in L_i, D_{ij} = |j - m(X_i)| \quad (4)$$

$$MAD_i = 1/n \sum_{k=1}^{n} D_{ij}, \text{ where n is the cardinality of } L_i \quad (5)$$

C. Identification using MAD plot method

For each user MAD_i of all N pages is plotted and compared with the mean of all pages. If the curve representing the MAD_i deviates more from the mean curve for all the pages then , it is concluded that the MAD curve is an outlier which falls into any of the above mentioned category of bots sending requests with

A) TSP equal to zero (DDoS bots)
B) Periodic or random TSP with much difference from mean TSP
C) Random TSP which blends into the real traffic.

Functionality	CLC,CC, Walrus	NC	Client
Nodes	Server#1	Server#{2,3,4}	Client#{1,2,3}
No. of NICs	eth0 - Enterprise N/W eth1- Eucalyptus N/W	eth0 -Eucalyptus N/W	eth0 - Enterprise N/W eth1 -Eucalyptus N/W
Subnet IP	eth0 - 10.1.60.0 eth1 - 10.1.50.0	eth0 - 10.1.50.0	eth0 - 10.1.60.0 eth1 - 10.1.50.0
VT enabled	No	Yes	Yes
CPU	64bit,3GHz,Multicore	64bit,3GHz,Multicore	64bit,3GHz,Multicore
Networking	100Mbps	100Mbps	100Mbps
Memory	4GB	4GB	4GB

Fig. 5. Eucalyptus Cloud Configuration

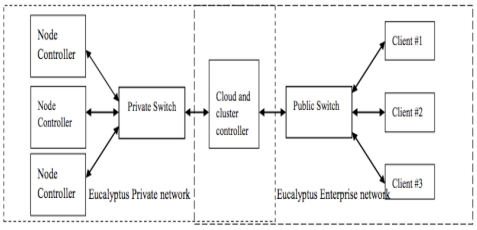


Fig. 6. Eucalyptus Network Setup

D. Identification using Foot Step graph method

Attack types A and C sometimes cannot be differentiated from the plot of MAD as the MAD in both cases do not deviate much from the mean. A foot step graph can identify the attack. Attack A can be identified if all the requests when plotted as a foot step graph represent a straight line indicating that TSP of all requests are nearly zero.Requests of type C are blended into the real traffic and are difficult to detect from the genuine traffic.

VI. Experimental Results

In order to carry out the experiment an eucalyptus cloud is set up with the configuration shown in Figure 5 and network setup shown in Figure 6. A web application is hosted on one of the node controllers and three clients act as bot machines, programmed to generate requests with characteristics of attack type A , B and C. The proposed detection mechanism is programmed into the cloud controller.

Normal traffic from other clients along with attack traffic from the bot machines is collected into a log file over a considerable period of time. The log is analyzed and a MAD plot is drawn for the three bot machines as shown in Figure 7. Client 1 is of type B, Client 2 is of type C and Client 3 is of type A. It is clearly observed from the plot that client 1 is an outlier.

To make further analysis a foot step graph is plotted as shown in Figure 8 . It can be observed that client 3 is of attack type A as its TSP plot represents a straight line indicating the requests with TSP equal to zero. Client 2 is of attack type C which looks much similar to the real traffic and becomes indifferentiable.

VII. Conclusion and Future work

The elastic nature of cloud provides a means to an attacker who targets the economy of a cloud consumer termed as economic denial of sustainability attack. The cloud consumer is billed for the intentionally created unprofitable traffic. In order to avoid unnecessary billing and creation of

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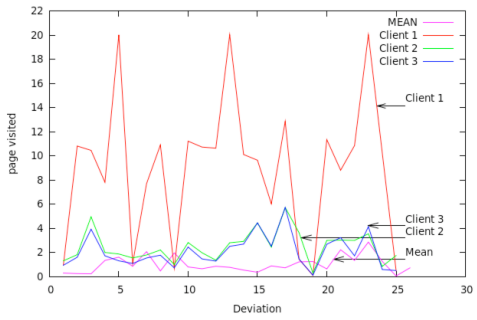


Fig. 7. MAD plot

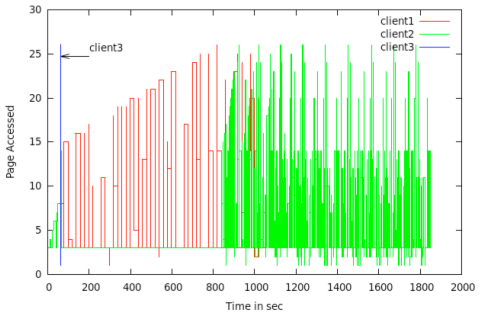


Fig. 8. Foot Step Plot

additional cloud resources there is a need to build a detection system that differentiates attack traffic from the genuine traffic.

TSP is identified as a mechanism for this differentiation. The attack and the genuine traffic differ in the TSP's of the requests they generate. This difference is calculated in terms of MAD. The mean and the MAD values are calculated. MAD plot method and Foot step Graph method plot the deviation and the TSP's respectively to identify the various types of attack traffic. This work can be further expanded to find out methods for interpreting the plots automatically instead of an administrator monitoring the graphs.

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